1. Check SSH availability: Run 'ssh' and write the output.

2. Verify GitHub SSH authentication: Run 'ssh -T git@github.com' and write the output.

3. List SSH files: Run 'ls -la ~/.ssh' and identify the Public Key and Private Key.

4. Generate a new SSH Key using 'ssh-keygen -t ed25519 -C "your_email@gmail.com"' and verify

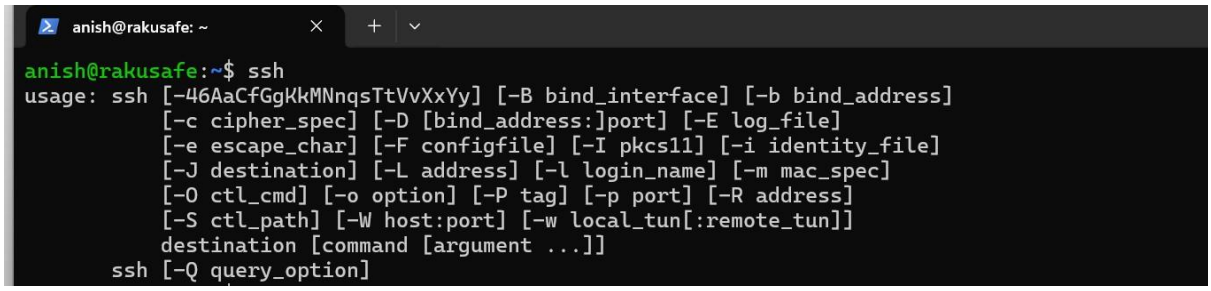
the files are created.

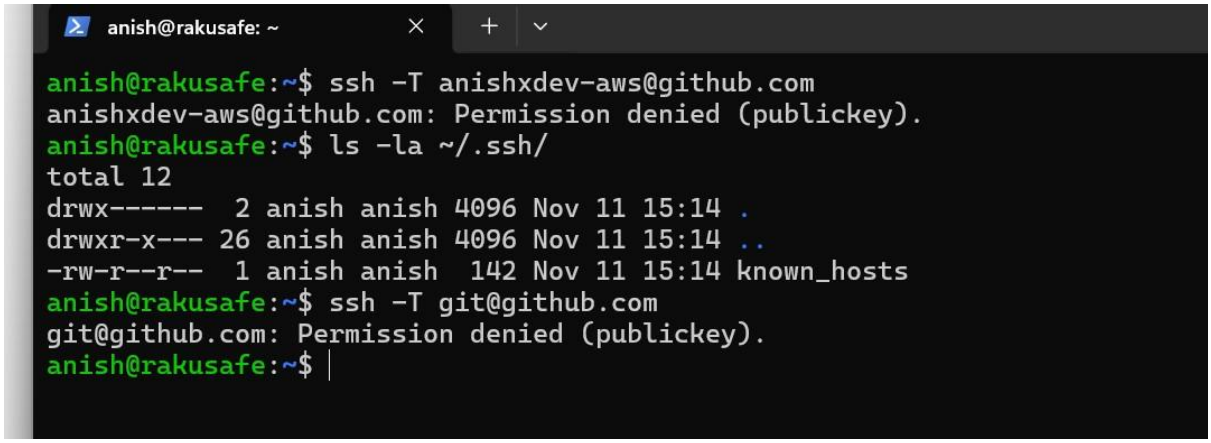
5. Start SSH Agent using 'eval "$(ssh-agent -s)"' and write the output.

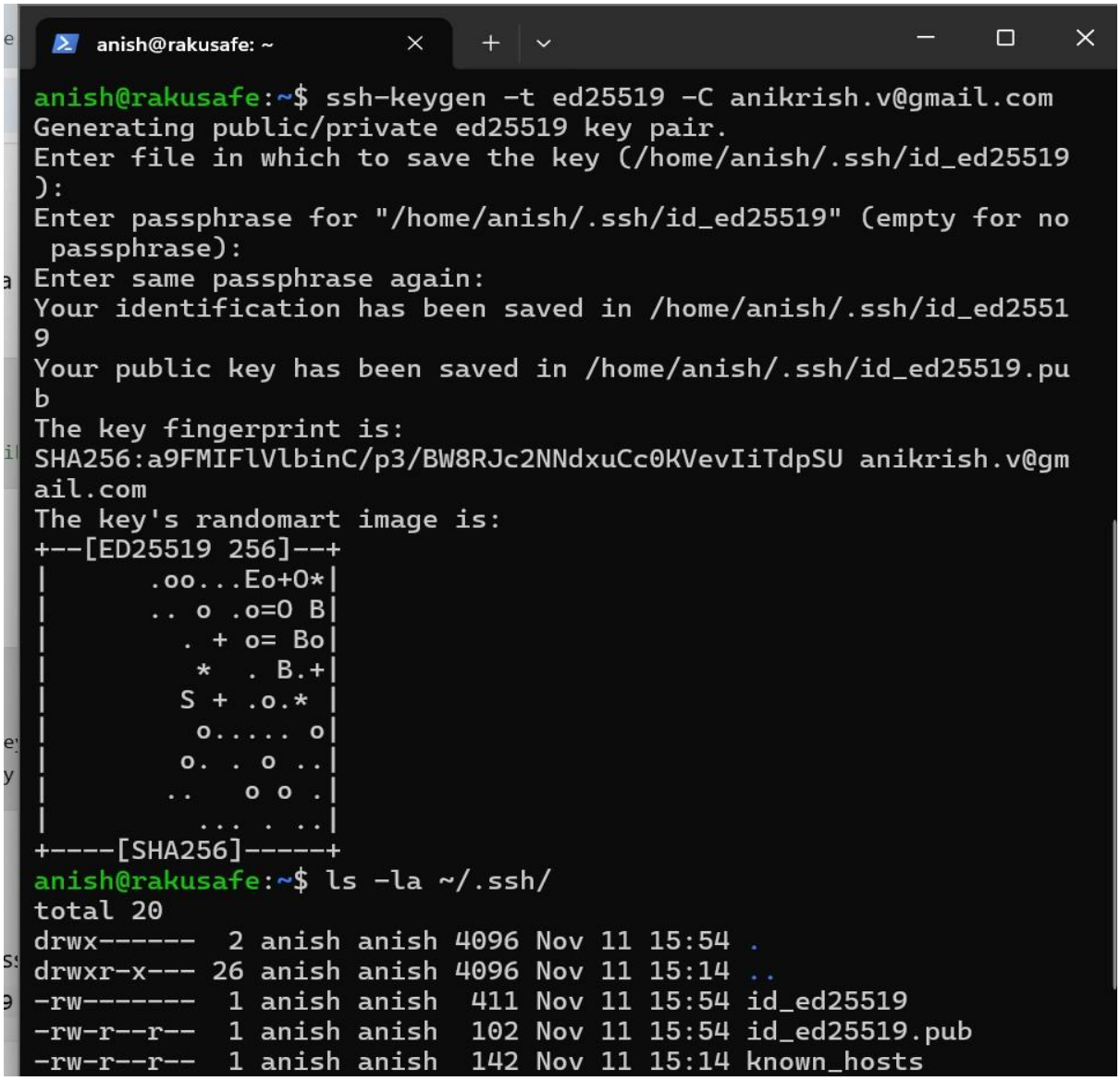
6. Add SSH Key to Agent using 'ssh-add ~/.ssh/id_ed25519' and write the output.

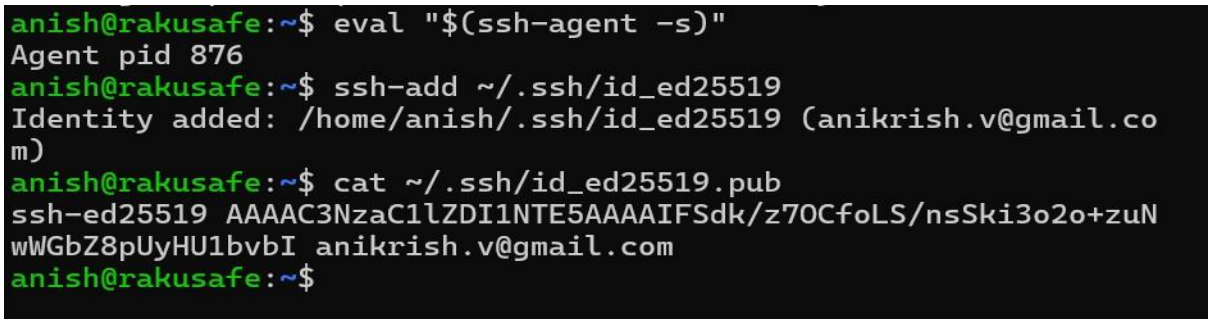
7. Display Public Key using 'cat ~/.ssh/id_ed25519.pub'. Copy the first few characters of the key.

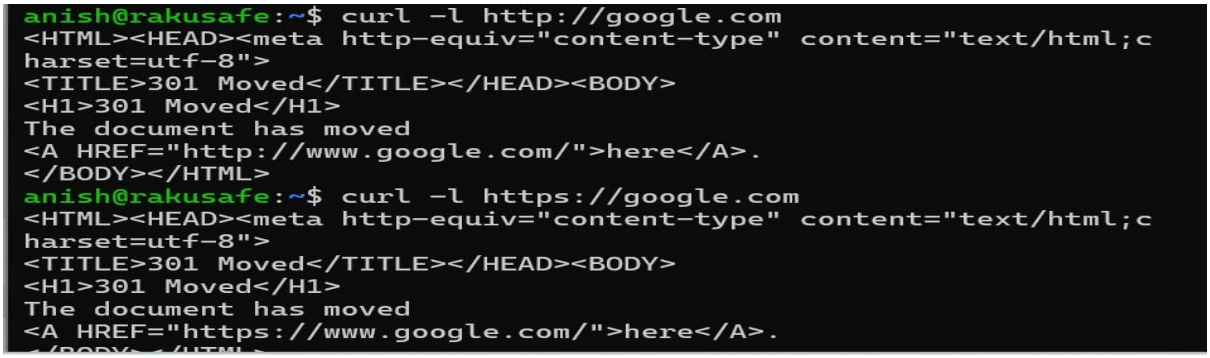
8. Check HTTPS Response using 'curl -I https://google.com'. What status code is returned?






anish@rakusafe: ~
anish@rakusafe:~$ ssh-keygen -t ed25519 -C anikrish.v@gmail.com
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/anish/.ssh/id_ed25519
):
Enter passphrase for "/home/anish/.ssh/id_ed25519" (empty for no
passphrase):
Enter same passphrase again:
Your identification has been saved in /home/anish/.ssh/id_ed2551
9
Your public key has been saved in /home/anish/.ssh/id_ed25519.pu
b
The key fingerprint is:
SHA256:a9FMIFlVlbinC/p3/BW8RJc2NNdxuCc0KVevIiTdpSU anikrish.v@gm
ail.com
The key's randomart image is:
+--[ED25519 256]--+
| .oo...Eo+O*|
| .. o .o=O B|
| . + o= Bo|
| * . B.+|
| S + .o.* |
| o..... o|
| o. . o ..|
| ..   o o .|
| ... . ..|
+----[SHA256]-----+
anish@rakusafe:~$ ls -la ~/.ssh/
total 20
drwx------ 2 anish anish 4096 Nov 11 15:54 .
drwxr-x--- 26 anish anish 4096 Nov 11 15:14 ..
-rw------- 1 anish anish 411 Nov 11 15:54 id_ed25519
-rw-r--r-- 1 anish anish 102 Nov 11 15:54 id_ed25519.pub
-rw-r--r-- 1 anish anish 142 Nov 11 15:14 known_hosts


anish@rakusafe:~$ eval "$(ssh-agent -s)"
Agent pid 876
anish@rakusafe:~$ ssh-add ~/.ssh/id_ed25519
Identity added: /home/anish/.ssh/id_ed25519 (anikrish.v@gmail.co
m)
anish@rakusafe:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFSdk/z7OCfoLS/nsSki3o2o+zuN
wWGbZ8pUyHU1bvbI anikrish.v@gmail.com
anish@rakusafe:~$



11. What is SSH?

SSH (Secure Shell) is a secure network protocol used to remotely connect to and manage another computer or server over a network.

Example:

ssh username@server_ip

12. Why is SSH required in DevOps?

SSH is required in DevOps because it allows us to:

1. Remotely access Linux servers.
2. Manage AWS EC2 instances.
3. Deploy applications on remote servers.
4. Transfer files securely.
5. Execute commands remotely.
6. Automate server administration tasks.

13. Which port does SSH use?

SSH uses TCP port 22 by default.
Example:
ssh -p 22 username@server_ip

14. What is SSH Key Authentication?

SSH Key Authentication is a secure login method that uses a pair of cryptographic keys instead of a password.

- Public Key – Stored on the remote server.
- Private Key – Stored securely on your local computer.

It allows users to connect to servers without entering a password every time.

15. Difference between Public Key and Private Key

Public Key	Private Key
Shared with the server.	Kept secret on your computer.
Stored in authorized_keys.	Stored in a .pem or other private key file.
Used to verify authentication.	Used to prove your identity.
Can be shared safely.	Must never be shared.

Example:

- mykey.pem – Private key.
- authorized_keys – Contains the public key on the server.

16. What is SSL?

- SSL (Secure Sockets Layer) is a security protocol designed to encrypt communication between a client and a server.
- It helps protect sensitive information such as passwords and payment details during transmission.
- Note: SSL is outdated and no longer considered secure.

17. What is TLS?

TLS (Transport Layer Security) is the modern security protocol used to encrypt communication between clients and servers.
It provides:

1. Data encryption.
2. Authentication.

3. Data integrity.
   Example: HTTPS websites use TLS to secure communication.

18. Difference between SSL and TLS

SSL	TLS
Stands for Secure Sockets Layer.	Stands for Transport Layer Security.
Older security protocol.	Newer security protocol.
Less secure.	More secure.
Deprecated and outdated.	Currently used.
SSL 2.0 and 3.0 are obsolete.	TLS 1.2 and 1.3 are widely used.

19. Why is HTTPS more secure than HTTP?

HTTPS (Hypertext Transfer Protocol Secure) is more secure than HTTP because it uses TLS encryption to protect communication between the browser and the web server.

Main advantages:

1. Encryption: Protects data from being read by attackers.
2. Authentication: Verifies the website's identity using a digital certificate.
3. Data Integrity: Prevents unauthorized modification of data during transmission.

20. Which port does HTTPS use?

HTTPS uses TCP port 443 by default.

- HTTP – Port 80
- HTTPS – Port 443
- SSH – Port 22

21. What is an SSL/TLS Certificate?

An SSL/TLS Certificate is a digital certificate used to verify a website's identity and enable encrypted communication between a browser and a web server.

It helps establish a secure HTTPS connection.

22. What information is stored inside a certificate?

An SSL/TLS certificate contains:

1. Domain Name: Website name (e.g., google.com).

2. Organization Name: Website owner's organization (if applicable).
3. Public Key: Used for secure communication.
4. Issuer Name: Certificate Authority (CA) that issued the certificate.
5. Validity Period: Certificate issue and expiry dates.
6. Digital Signature: Verifies the certificate's authenticity.
7. Serial Number: Unique certificate identification number.

23. What does the 'Connection is Secure' message in a browser indicate?

“Connection is Secure” in a browser means that the website is using HTTPS with a valid TLS certificate to establish an encrypted connection between your browser and the web server.

It indicates:

1. Encryption: Data exchanged between browser and server is encrypted.
2. Authentication: The certificate helps verify the website's identity.
3. Data Integrity: Data cannot be modified without detection during transmission.

Example: When you visit https://google.com, the browser establishes a secure HTTPS connection.

Important: A secure connection does not guarantee that the website itself is genuine or trustworthy. It only indicates that the connection is secured.

Interview answer (short):

"Connection is Secure means the browser has established an encrypted HTTPS connection using TLS, helping protect data during transmission and verify the website's identity."

24. Explain the SSL/TLS Handshake flow: Client Hello, Server Hello, Certificate, Verification, Key

Exchange, Encrypted Session.

"TLS Handshake is the process of establishing a secure connection between a client and server. First, the client sends Client Hello. The

server responds with Server Hello and its certificate. The client verifies the certificate. Then both exchange key material to establish shared session keys. Finally, encrypted communication begins."

Remember these six steps:

Client Hello → Server Hello → Certificate → Verification → Key Exchange → Encrypted Session.

25. Explain the production flow: User → DNS → HTTPS → Load Balancer → Nginx → Docker Container → Database → Response.

Production Flow explains how a user accesses an application hosted on a production server and how the request travels through different infrastructure components before receiving a response.

User
Requests the application through a browser.
DNS
Converts the domain name into an IP address.
HTTPS
Provides secure communication using TLS.
Load Balancer
Distributes traffic among multiple servers.
Nginx
Forwards requests to the application.
Docker Container
Runs the application in an isolated environment.
Database
Stores application information.
Response
Sends the requested result back to the user.

26. Explain the output of curl -I https://google.com.

Answer:

The ***curl -I https://google.com*** command is used to retrieve only the HTTP response headers from Google's server without downloading the complete webpage content.

curl -I https://google.com

output:-
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: ...

Important observations:

1. HTTP does not establish a TLS connection.
2. HTTPS establishes a secure TLS connection.
3. Both can return HTTP status codes such as 200, 301 and 404.
4. Google may redirect HTTP requests to HTTPS or another URL.

28. List three real-world uses of SSH in DevOps.

SSH (Secure Shell) is used to securely access and manage remote servers.
Three real-world uses:

1. Remote Server Management: Administrators use SSH to connect to Linux servers and execute commands remotely.
2. Application Deployment: DevOps engineers use SSH to deploy applications and restart services on remote servers.
3. Secure File Transfer: SSH is used with SCP or SFTP to transfer files securely between computers and servers.

9. Compare HTTP and HTTPS using curl. Write two differences.

Commands:
curl -I http://google.com
curl -I https://google.com
Two differences:

HTTP	**HTTPS**
Uses port 80 by default.	Uses port 443 by default.
Does not use TLS encryption.	Uses TLS encryption to secure communication.

Conclusion: HTTPS is more secure than HTTP because it provides encryption, authentication and data integrity.

10. Scenario Task: AWS EC2 IP = 54.234.xxx.xxx, key file = mykey.pem. Write the SSH command used to connect.

Answer:
ssh -i mykey.pem ubuntu@54.234.xxx.xxx
Explanation:

- ssh – Secure Shell command.
- -i – Specifies the private key file.
- mykey.pem – Private key filename.
- ubuntu – EC2 username (assuming Ubuntu OS).
- 54.234.xxx.xxx – EC2 public IP address.